ISDB-BISEW

IT Scholarship Programme

Round 73

DATABASE DESIGN PROJECT

Jobs Portal

A MySQL database for a job website where employers post jobs and job seekers apply.

PHP MySQL HTML CSS JavaScript

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PROJECT IDEA

The problem it solves

The problem

- Job seekers look for jobs on many different sites and pages
- Employers find it hard to reach the right people
- Applying by email or paper is slow and hard to track

My solution

- One website where all jobs are in one place
- Employers post jobs and see every applicant
- Seekers apply online and see their status any time

Goal: make finding a job and hiring a person simple, fast and clear.

THE PROJECT

Three types of users

Job Seeker

- Builds a CV with the CV builder
- Searches and saves jobs
- Applies and tracks the status

Employer

- Creates a company profile
- Buys a package and posts jobs
- Shortlists and hires applicants

Admin

- Approves new jobs
- Checks and approves payments
- Manages users, categories and CV templates

PROJECT IDEA

How it works in three steps

Job seeker

1 Create a free account → 2 Build a CV in the site or upload one → 3 Apply and follow the status

Employer

1 Register the company → 2 Choose a package and post a job → 3 Shortlist and hire the best people

PROJECT IDEA

Finding the right job is easy

Job title, skill or company | All categories | All cities | Search

Filter by: Category | City | Job type | Experience | Salary | Remote

☆ Featured jobs are shown first ☐ Save a job and apply later ◷ Old jobs hide after the deadline

REFERENCE WEBSITE

Based on a real job portal

Home
Search, categories, jobs

Jobs
All jobs with filters

Companies
Company profiles

Search Talent
Employers find seekers

Login / Register
Seeker or company

Contact
Questions to the site

I studied the main pages of a demo job portal, then designed my own simpler database for the same features.

DATABASE STRUCTURE

20 tables in 7 groups

Users & profiles
- users
- companies
- seekers

Lookup lists
- cities
- categories
- skills
- cv_templates

Seeker CV
- seeker_educations
- seeker_experiences
- seeker_skills
- seeker_cv_items

Jobs
- jobs
- job_skills

Applying
- job_applications
- saved_jobs

Packages
- packages
- subscriptions

Payments & alerts
- payment_methods
- payment_transactions
- notifications

20
tables in total

CV BUILDER

CV builder: make a CV inside the site

1

Fill the profile

Name, city, category, headline and a short summary

`users`
`seekers`

2

Education & jobs

Add each degree and each past or current job

`seeker_educations`
`seeker_experiences`

3

Skills & more

Pick skills, then add projects, certificates and languages

`seeker_skills`
`seeker_cv_items`

4

Design & download

Choose a CV design, save the PDF and apply with it

`cv_templates`
`seekers.resume_file`

FROM TABLES TO CV

Every part of the CV comes from a table

Rahim Ahmed

Junior PHP Developer · Dhaka · rahim@example.com

SUMMARY — `seekers`

Web developer with 2 years of experience in PHP and MySQL.

EDUCATION — `seeker_educations`

BSc in CSE · Dhaka University · 2023

EXPERIENCE — `seeker_experiences`

Web Developer · ABC Soft Ltd · 2024 – now

SKILLS — `seeker_skills → skills`

PHP · MySQL · Laravel · JavaScript

PROJECTS — `seeker_cv_items · type = project`

Jobs Portal · PHP, MySQL job website

CERTIFICATES — `seeker_cv_items · type = certificate`

Web Application Development · IsDB-BISEW · 2026

LANGUAGES — `seeker_cv_items · type = language`

Bangla (native) · English (good)

One table, many CV parts

Projects, certificates and languages all go in **seeker_cv_items**. The **type** column says which one it is, so 3 tables become 1.

Designs from the admin

The admin adds CV designs in **cv_templates**. The seeker's choice is saved in **seekers.cv_template_id**.

The CV goes with each application

When a seeker applies, the PDF is copied to **job_applications.resume_file**, so the employer sees the CV as it was that day.

Sample data shown. The CV builder adds only 2 tables: **cv_templates** and **seeker_cv_items**.

GROUP 1

Users and profiles

Table	What it stores	Linked to
users	Login details of every user: admin, employer or seeker	–
companies	The company profile of an employer: name, logo, address	users, cities
seekers	The profile of a job seeker: headline, summary, CV design and CV file	users, cities, categories, cv_templates

Key idea: everyone logs in from one **users** table. The **role** column says who they are, and their details are kept in **companies** or **seekers**.

GROUPS 2 AND 3

Lookup lists and the seeker's CV

Table	What it stores	Linked to
cities	Cities with their division	companies, seekers, jobs
categories	Job categories, such as Sales or IT	jobs, seekers
skills	Skill names, such as PHP or Excel	jobs, seekers
cv_templates	CV designs a seeker can choose	seekers
seeker_educations	Degrees and results of a seeker	seekers
seeker_experiences	Past and current jobs of a seeker	seekers
seeker_skills	Which skills each seeker has	seekers, skills
seeker_cv_items	Projects, certificates and languages, marked by a **type** column	seekers

The first four are lists the admin manages. The last four hold the CV details of a seeker.

GROUPS 4 AND 5

Jobs and applications

Table	What it stores	Linked to
jobs	Job posts: title, salary, deadline and status	companies, categories, cities
job_skills	The skills a job needs	jobs, skills
job_applications	Each application, the CV sent and its status	jobs, seekers
saved_jobs	Jobs a seeker has bookmarked	jobs, seekers

Application status: applied · viewed · shortlisted · interview · hired · rejected

GROUPS 6 AND 7

Packages, payments and notifications

Table	What it stores	Linked to
packages	Plans to buy: Free, Standard, Premium	–
subscriptions	A bought package and the job posts left	users, packages
payment_methods	Cash, bKash, Nagad, Rocket, Bank, Card	–
payment_transactions	Every payment: amount, TrxID, status	users, subscriptions, payment_methods
notifications	Messages like “You are shortlisted”	users

RELATIONSHIPS

Three ways the tables connect

One to one

`users → companies`
`users → seekers`

One employer has exactly one company profile.

One to many

`companies → jobs`
`jobs → job_applications`

One company can post many jobs. One job gets many applications.

Many to many

`jobs ↔ skills`
`via job_skills`

A job needs many skills, and a skill is used in many jobs. A middle table joins them.

Tables are joined with a **foreign key**: a column that stores the **id** of a row in another table, such as **jobs.company_id**.

PAYMENT FLOW

How a package is bought

1

Choose a package

A new row in **subscriptions**

pending

→

2

Pay and send the TrxID

A new row in **payment_transactions**

pending

→

3

Admin approves

Payment and package both become ready

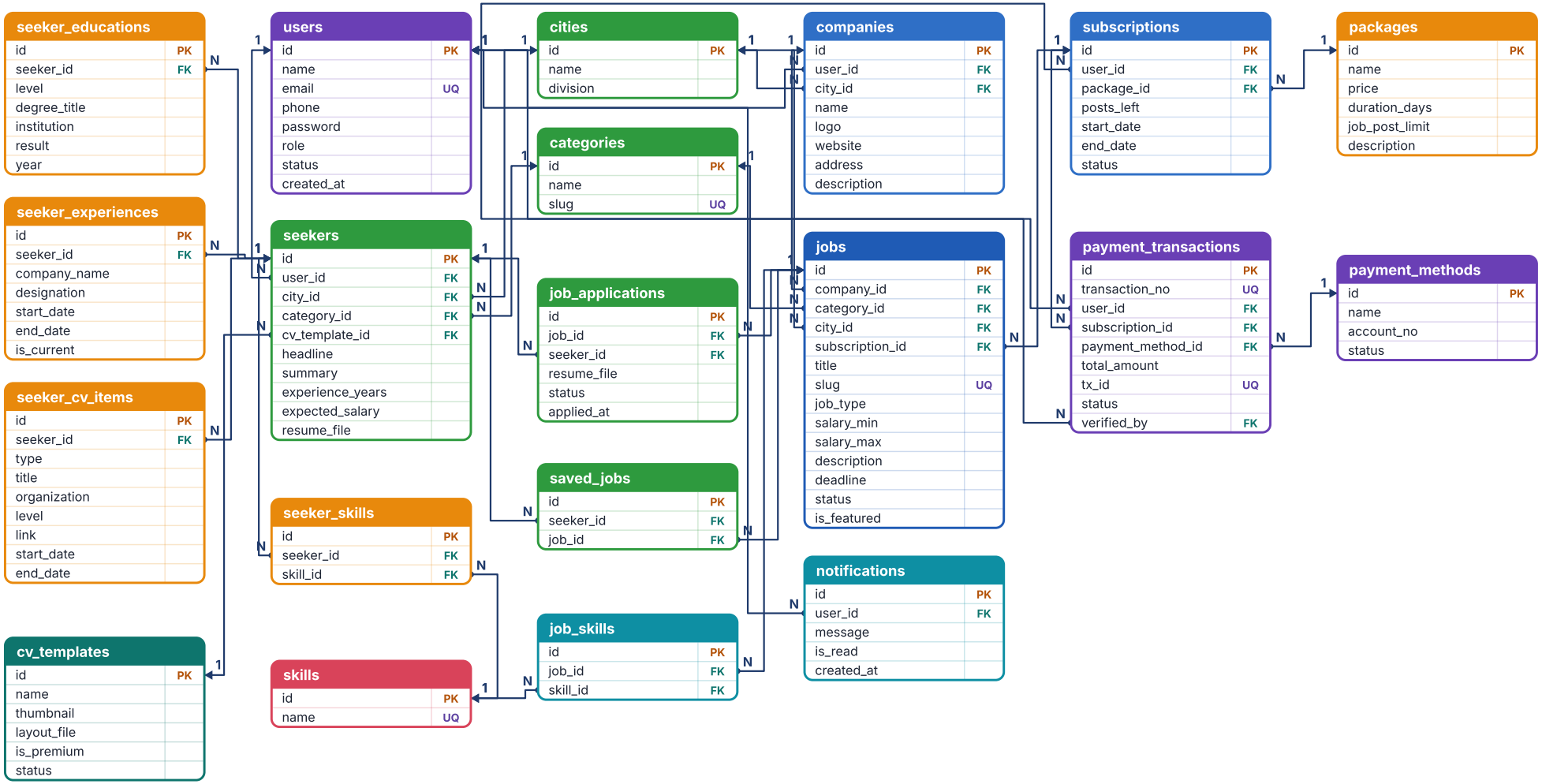
completed

active

Each payment is linked to one row in **payment_methods**: Cash, bKash, Nagad, Rocket, Bank Transfer or Card.

DATABASE DIAGRAM

ERD: all 20 tables



PK = primary key, **FK** = foreign key, **UQ** = unique. Each line goes from a foreign key (**N** side) to the **id** it points to (**1** side).

PROJECT IDEA

Ideas for the future

Email job alerts

Send new matching jobs to seekers by email

Chat

Employers and seekers send messages inside the site

Mobile app

Search and apply from an Android or iPhone app

Bangla and English

Let users choose the language of the website

SUMMARY

Simple, linked and safe

20 tables in 7 groups

3 user roles in one login table

6 payment methods

- ✓ A seeker can apply to a job only once
- ✓ Passwords are saved as secure hashes
- ✓ The same TrxID cannot be used twice
- ✓ Deleting a user also removes their profile